

EC-350 AI and Decision Support Systems

Week 15 Decision Trees

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Introduction to Decision Trees

- Metric methods do not work with nominal features (the ones that do not have a similarity measure or ordering)
- Feature vector is replaced by a list of attributes (might not even have the same number of features sometimes):
- Examples:
 - *Fruits: (size, color, shape, taste) (small, yellow, round, sour)*
 - *Weather: (clouds, temperature, wind, humidity) (cloudy, cold, calm, high)*
 - *Credit: (has, amount, salary, debt, paid before) (yes, large, large, yes, no)*

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Decision Trees

- Classify the pattern using a sequence of questions in which the next question depends on the answer to the previous question
- This approach allows for non-metric data use
- The graphical representation is a tree where
 - Each non leaf node corresponds to a question
 - The leaves are associated with categories
 - To classify, start from the root and follow the path generated by the answers.

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Decision Trees

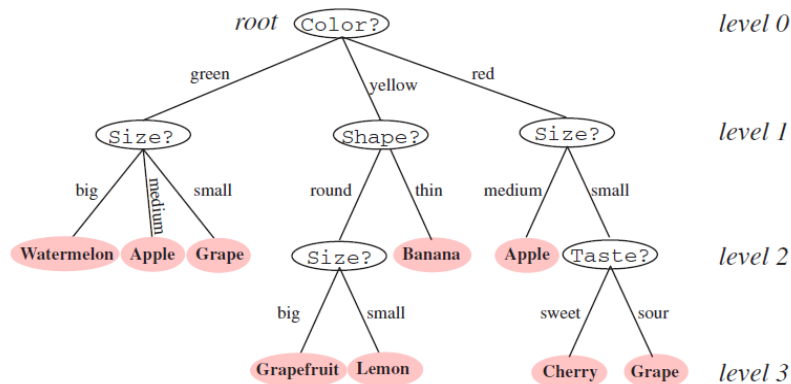


Figure 8.1: Classification in a basic decision tree proceeds from top to bottom. The questions asked at each node concern a particular property of the pattern, and the downward links correspond to the possible values. Successive nodes are visited until a terminal or leaf node is reached, where the category label is read. Note that the same question, *Size?*, appears in different places in the tree, and that different questions can have different numbers of branches. Moreover, different leaf nodes, shown in pink, can be labeled by the same category (e.g., *Apple*).

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Classification Decision Trees

- **Nonmetric.** Data is list of binary attributes.
- Apple is *red* and *round*. Banana is *long* and *yellow*.
- Difficult to apply *nearest neighbor* and other techniques.

- **Decision Tree:** based on game of twenty questions.
- Apply a series of tests to the input pattern
- Each test asks a question: e.g. “is the pattern yellow?”
- The answer is “yes” or “no”.
 - The answers give the classification -- e.g. the pattern is yellow, not-round, and long – so it is a banana.

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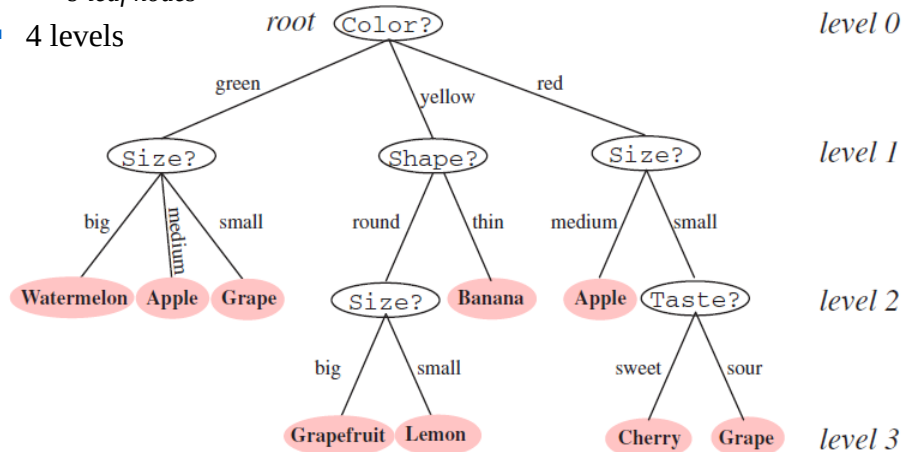
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Decision Tree Example

- 15 nodes:
 - 1 root node
 - 9 leaf nodes
- 4 levels



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CART: Learn Decision Tree

- CART is a general framework for designing decision trees.
- Basic Issues:
 - (1) Should we restrict ourselves to binary questions?
 - (2) Which attributes should be tested at each node?
 - (3) When should a node be declared a leaf?
 - (4) How can we prune a large tree?
 - (5) How do we assign a class label to a leaf node?

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Number of Splits - Binary

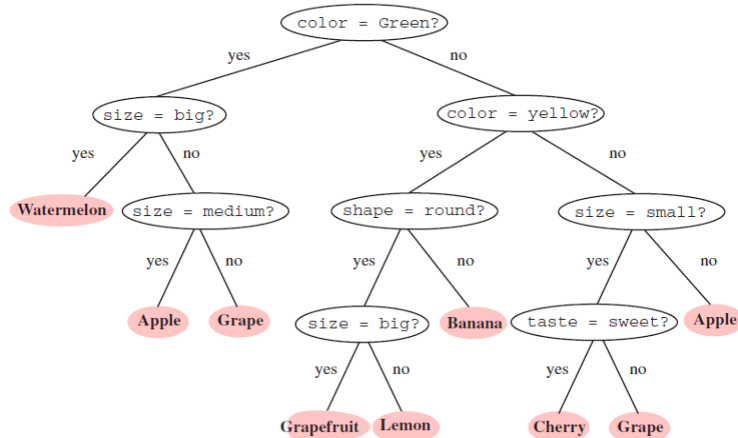


Figure 8.2: A tree with arbitrary branching factor at different nodes can always be represented by a functionally equivalent binary tree, i.e., one having branching factor $B = 2$ throughout. By convention the “yes” branch is on the left, the “no” branch on the right. This binary tree contains the same information and implements the same classification as that in Fig. 8.1.

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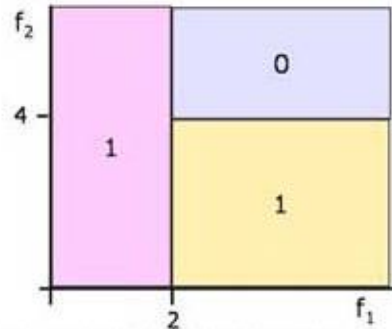
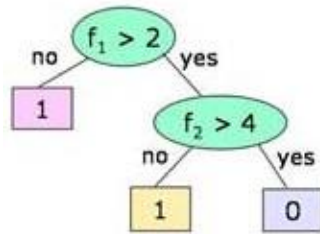
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Decision Boundaries

- Tests in nodes can be of the form $x_i \leq \text{constant}$
- Divides the space into axis-aligned rectangles.



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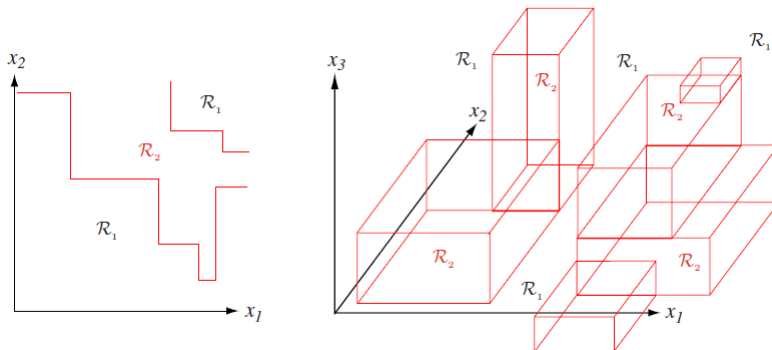
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Queries and Decision Boundaries

- Generally queries for numeric data are given in the form $x_i \leq x_{is}$
- Such queries result in decision boundaries shown in the figure:



- A basic question will be how should the query be selected?

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Query Selection: Impurity

- Iterative Design Principle: For all leaf nodes, find (by searching over all tests) the test that maximizes the classification (based on information theory measures such as entropy).
- Expand the nodes and add its child nodes to the tree.
- Impurity as a measure of quality/classification of test
 - Best test makes the children node as 'pure' as possible
 - Decrease in impurity: The lower the impurity, the easier to classify.

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Impurity Measures

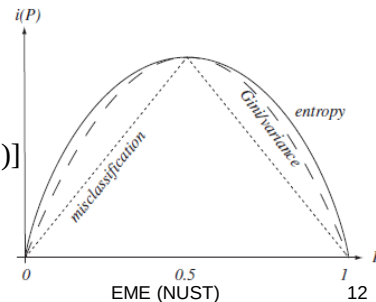
- $i(N)$ – impurity at node N
- **Entropy impurity:**

$$i(N) = - \sum_{i \in c} P(\omega_i) \log_2 P(\omega_i)$$
 - $P(\omega_i)$ is the fraction of patterns at node N that are in category ω_i
 - Is large when the classes are equally likely, zero when only one class is possible
- **Variance impurity: 2 class case:**

$$i(N) = P(\omega_1)P(\omega_2)$$
- **Gini Impurity:**

$$i(N) = \sum_{i \neq j} P(\omega_i)P(\omega_j) = \frac{1}{2} [1 - \sum_i P^2(\omega_i)]$$
- **Misclassification impurity**

$$i(N) = 1 - \max P(\omega_i)$$



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Query Selection Criterion: Decrease in Impurity

- The decrease in impurity after a query at node N is:

$$\Delta i(N) = i(N) - (P_L i(N_L) + (1 - P_L) i(N_R))$$

- N_L, N_R are the left and the right children nodes of N
- P_L is the fraction of patterns at node N that will go to the left child node
- Query selection criterion
 - If T represents the query and s represents the query value (decision), the best query value s is the choice for maximum $\Delta i(T)$

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Example

Color	Size	Taste	Fruit
Yellow	Big	Sweet	Apple
Yellow	Big	Sweet	Apple
Red	Small	Sour	Orange
Red	Small	Sour	Orange
Red	Small	Sour	Orange
Red	Big	Sour	Orange
Red	Big	Sweet	Orange
Yellow	Small	Sour	Apple
Yellow	Big	Sour	Apple
Yellow	Big	Sour	Apple
Yellow	Small	Sweet	Orange

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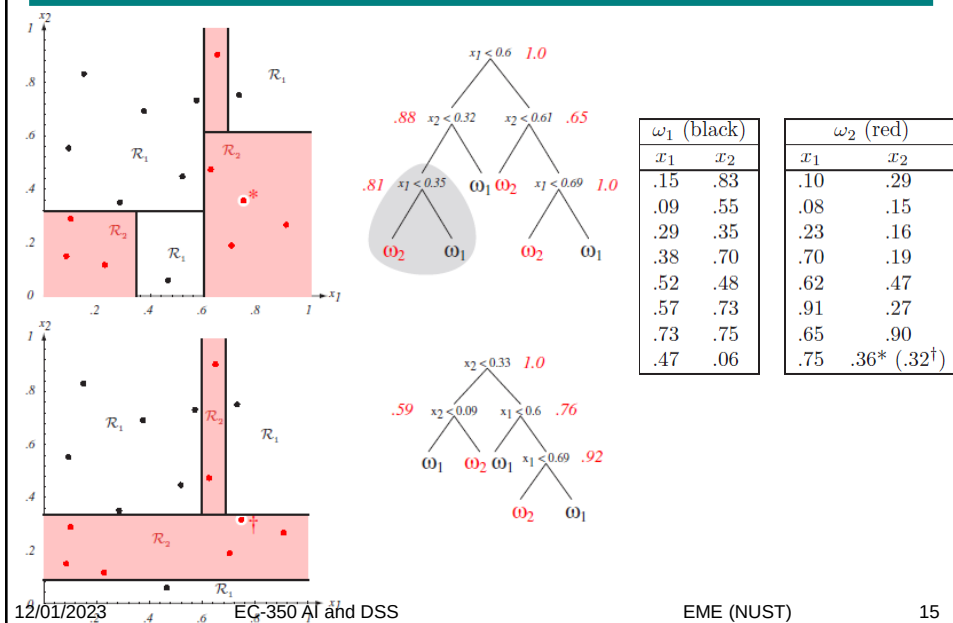
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Decision Trees: Example



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Learning Decision Tree

- Greedy Strategy:
 - Start at root node. Select the test that has maximum impurity decrease
 - For each child node, select the test that has maximal impurity decrease
 - Repeat until each node is a pure leaf node, $i(N) = 0$
 - Classify test data by just passing through the tree
- Algorithm is greedy. There may be a more efficient tree (fewer tests) if you learned the tree by searching over the sequence of tests.
- The choice of number of children are arbitrary. There might be a more efficient tree (fewer tests) if the tests were multi-valued.

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